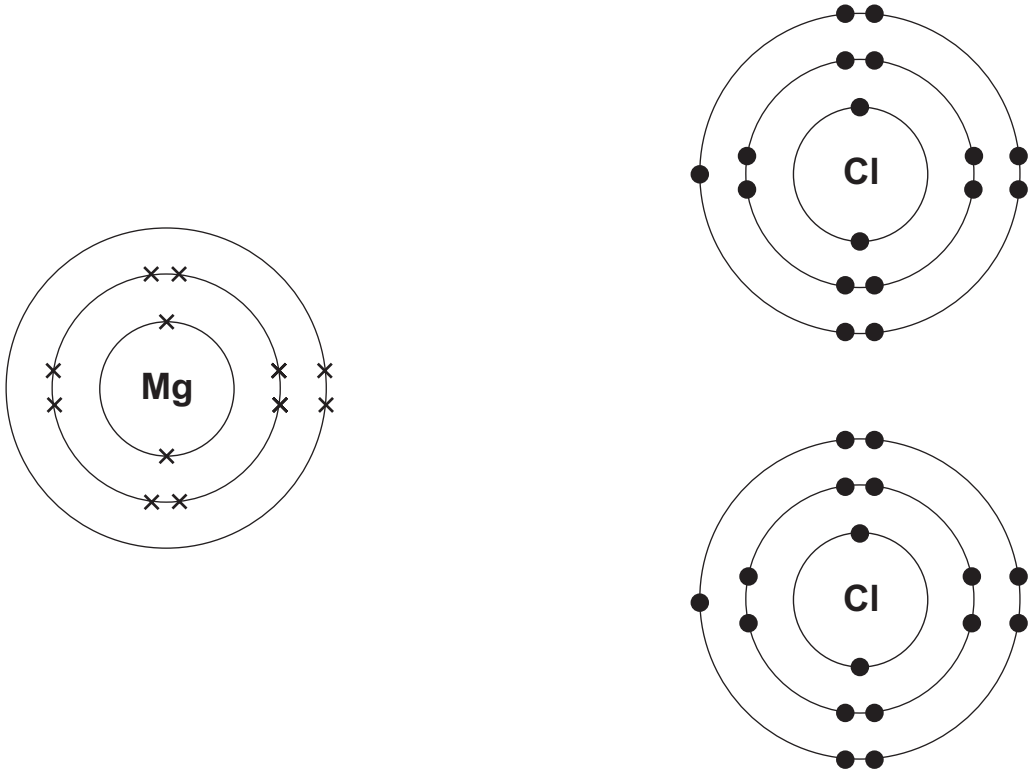


3. (a) Magnesium reacts with chlorine to form magnesium chloride.

The following diagram shows the electronic structures of magnesium and chlorine atoms.



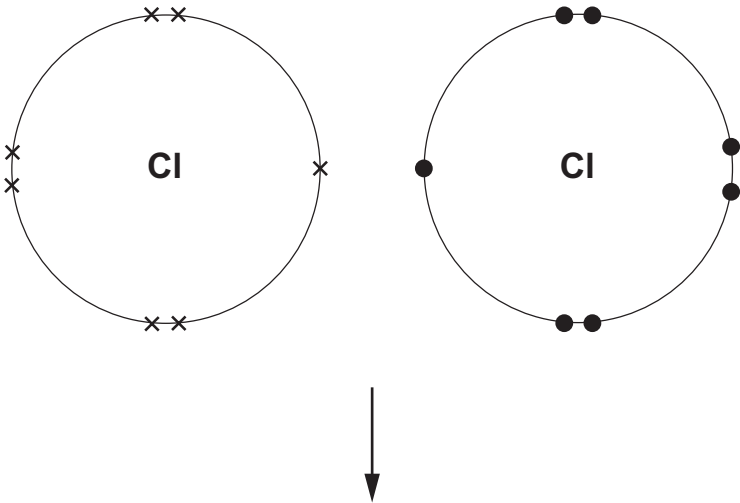
- (i) **Draw arrows on the diagram** to show how electrons are transferred between the magnesium and chlorine atoms during the formation of magnesium chloride. [1]
- (ii) **Complete the table** giving information about the charge and electronic structure of the magnesium and chloride **ions** that are formed. [2]

Ion	Charge	Electronic structure
magnesium	+2	
chloride		2,8,8



(b) Chlorine gas, Cl_2 , consists of two chlorine atoms bonded together.

(i) Draw a diagram in the box to show how the atoms bond to form a chlorine molecule. [1]



(ii) Give the name of this type of bonding.

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[1]

5

